

CHAPTER 5 : MASS AND ENERGY ANALYSIS OF CONTROL VOLUMES

Section	Core Revision Focus
5.1	Conservation of mass for control volumes; mass and volume flow rates, normal velocity, general mass balance, steady-flow mass balance, and the incompressible-flow continuity relation.
5.2	Flow work and the energy carried by flowing fluid; $w_{\text{flow}} = Pv$, enthalpy $h = u + Pv$, and total flowing-fluid energy $h + \frac{V^2}{2} + gz$.
5.3	Steady-flow energy analysis; mass and energy balances, the steady-flow energy equation, sign conventions, and identification of negligible kinetic, potential, heat-transfer, and work terms.
5.4	Application of mass and energy balances to steady-flow engineering devices: nozzles, diffusers, turbines, compressors, pumps, fans, throttling valves, mixing chambers, heat exchangers, and pipe/duct flow.
5.5	Unsteady-flow energy analysis; mass and energy accumulation, uniform-flow processes, distinction between flowing-fluid enthalpy and stored-fluid internal energy, and charging/discharging of tanks.

5.1 CONSERVATION OF MASS

MASS AND VOLUME FLOW RATES

For flow through a control surface, only the velocity component normal to the area contributes.

$$\delta \dot{m} = \rho V_n dA_c$$

$$\dot{m} = \int_{A_c} \rho V_n dA_c$$

$$V_{avg} = \frac{1}{A_c} \int_{A_c} V_n dA_c$$

For uniform/average velocity:

$$\dot{m} = \rho V_{avg} A_c$$

$$\dot{V} = \int_{A_c} V_n dA_c = V_{avg} A_c = V A_c$$

$$\dot{m} = \rho \dot{V} = \frac{\dot{V}}{v}$$

$$m = \rho V = \frac{V}{v}$$

$$\rho = \frac{1}{v}$$

CONSERVATION OF MASS PRINCIPLE

$$\text{Mass entering} - \text{Mass leaving} = \text{Change in mass within CV}$$

$$m_{in} - m_{out} = \Delta m_{CV}$$

$$\Delta m_{CV} = m_{final} - m_{initial}$$

Rate form:

$$\dot{m}_{in} - \dot{m}_{out} = \frac{dm_{CV}}{dt}$$

Mass contained in the control volume:

$$dm = \rho dV$$

$$m_{CV} = \int_{CV} \rho dV$$

$$\frac{dm_{CV}}{dt} = \frac{d}{dt} \int_{CV} \rho dV$$

For constant mass inside the CV:

$$\boxed{\frac{dm_{CV}}{dt} = 0}$$

NORMAL COMPONENT OF VELOCITY

$$V_n = V \cos \theta$$

$$\boxed{V_n = \vec{V} \cdot \vec{n}}$$

With outward unit normal:

$$V_n > 0 \quad \text{outflow} \quad V_n < 0 \quad \text{inflow}$$

$$\delta \dot{m} = \rho V_n dA = \rho (\vec{V} \cdot \vec{n}) dA$$

$$\boxed{\dot{m}_{net} = \int_{CS} \rho (\vec{V} \cdot \vec{n}) dA}$$

GENERAL CONSERVATION OF MASS EQUATION

$$\boxed{\frac{dm_{CV}}{dt} + \dot{m}_{out} - \dot{m}_{in} = 0}$$

Equivalent form:

$$\boxed{\frac{dm_{CV}}{dt} = \sum_{in} \dot{m} - \sum_{out} \dot{m}}$$

Integral form:

$$\boxed{\frac{d}{dt} \int_{CV} \rho dV = \sum_{in} \dot{m} - \sum_{out} \dot{m}}$$

If velocity is normal to the surface:

$$\vec{V} \cdot \vec{n} = V$$

$$\int_A \rho (\vec{V} \cdot \vec{n}) dA = \rho V A$$

MASS BALANCE FOR STEADY-FLOW PROCESSES

Steady flow:

$$m_{CV} = \text{constant} \quad \Rightarrow \quad \frac{dm_{CV}}{dt} = 0$$

Hence,

$$\sum_{\text{in}} \dot{m} = \sum_{\text{out}} \dot{m}$$

For a single inlet and outlet:

$$\dot{m}_1 = \dot{m}_2$$

$$\dot{m} = \rho VA$$

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

SPECIAL CASE: INCOMPRESSIBLE FLOW

For constant density:

$$\sum_{\text{in}} \dot{V} = \sum_{\text{out}} \dot{V}$$

For one inlet and one outlet:

$$\dot{V}_1 = \dot{V}_2$$

$$V_1 A_1 = V_2 A_2$$

For incompressible steady flow:

$$\dot{m}_{\text{in}} = \dot{m}_{\text{out}}$$

but, in general,

$$\dot{V}_{\text{in}} \neq \dot{V}_{\text{out}}$$

when density changes.

5.2 FLOW WORK AND THE ENERGY OF A FLOWING FLUID

FLOW WORK

Pressure force at an inlet/exit:

$$F = PA$$

Flow work:

$$W_{\text{flow}} = FL = PAL = PV$$

Specific flow work:

$$w_{\text{flow}} = Pv$$

Thus the pressure/flow energy carried with each unit mass is Pv .

TOTAL ENERGY OF A FLOWING FLUID

Specific total energy of a nonflowing fluid:

$$e = u + ke + pe$$

$$e = u + \frac{V^2}{2} + gz$$

For flowing fluid, include flow work:

$$\theta = Pv + e$$

$$\theta = Pv + u + \frac{V^2}{2} + gz$$

Define enthalpy:

$$h = u + Pv$$

Therefore,

$$\theta = h + \frac{V^2}{2} + gz$$

$$\boxed{\text{Flowing fluid: } h \quad \text{Nonflowing fluid: } u}$$

ENERGY TRANSPORT BY MASS

Energy transported with mass:

$$E_{\text{mass}} = m\theta$$

$$E_{\text{mass}} = m \left(h + \frac{V^2}{2} + gz \right)$$

Rate of energy transport:

$$\dot{E}_{\text{mass}} = \dot{m}\theta = \dot{m} \left(h + \frac{V^2}{2} + gz \right)$$

When kinetic and potential energies are negligible:

$$E_{\text{mass}} \approx mh \quad \dot{E}_{\text{mass}} \approx \dot{m}h$$

ENERGY TRANSPORT THROUGH AN INLET OR EXIT

For an infinitesimal mass:

$$\delta E_{\text{mass}} = \theta \delta m$$

$$E_{\text{in, mass}} = \int_{m_i} \theta_i \delta m_i$$

$$E_{\text{in, mass}} = \int_{m_i} \left(h_i + \frac{V_i^2}{2} + gz_i \right) \delta m_i$$

5.3 ENERGY ANALYSIS OF STEADY-FLOW SYSTEMS

CHARACTERISTICS OF A STEADY-FLOW PROCESS

For a steady-flow control volume:

$$V_{CV} = \text{constant}$$

$$m_{CV} = \text{constant}$$

$$E_{CV} = \text{constant}$$

Therefore, there is no accumulation of mass or energy within the CV.

MASS BALANCE FOR STEADY-FLOW SYSTEMS

$$\sum_{\text{in}} \dot{m} = \sum_{\text{out}} \dot{m}$$

Single stream:

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{m} = \rho VA$$

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

ENERGY BALANCE FOR STEADY-FLOW SYSTEMS

Since E_{CV} is constant,

$$\Delta E_{CV} = 0$$

Thus,

$$\dot{E}_{\text{in}} = \dot{E}_{\text{out}}$$

Using the sign convention of the chapter,

$$\dot{Q}_{\text{in}} + \dot{W}_{\text{in}} + \sum_{\text{in}} \dot{m}\theta = \dot{Q}_{\text{out}} + \dot{W}_{\text{out}} + \sum_{\text{out}} \dot{m}\theta$$

where

$$\theta = h + \frac{V^2}{2} + gz$$

Hence,

$$\dot{Q} - \dot{W} = \sum_{\text{out}} \dot{m} \left(h + \frac{V^2}{2} + gz \right) - \sum_{\text{in}} \dot{m} \left(h + \frac{V^2}{2} + gz \right)$$

Here,

$$\dot{Q} = \dot{Q}_{\text{net,in}}$$

$$\dot{W} = \dot{W}_{\text{net,out}}$$

SINGLE-STREAM STEADY-FLOW ENERGY EQUATION

For one inlet and one outlet:

$$\dot{m}_1 = \dot{m}_2 = \dot{m}$$

$$\dot{Q} - \dot{W} = \dot{m} \left[h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right]$$

Per unit mass:

$$q - w = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$

$$q = \frac{\dot{Q}}{\dot{m}} \quad w = \frac{\dot{W}}{\dot{m}}$$

NEGLECTIBLE KINETIC AND POTENTIAL ENERGY CHANGES

If

$$\Delta ke \approx 0 \quad \Delta pe \approx 0$$

then

$$q - w = h_2 - h_1$$

HEAT TRANSFER

$$\dot{Q} > 0 \quad \text{heat enters CV}$$

$$\dot{Q} < 0 \quad \text{heat leaves CV}$$

$$\dot{Q} = 0 \quad \text{adiabatic}$$

WORK TRANSFER

For no work interaction:

$$\dot{W} = 0$$

ENTHALPY CHANGE

$$\Delta h = h_2 - h_1$$

For an ideal-gas-like constant- c_p approximation:

$$\Delta h = c_{p,avg}(T_2 - T_1)$$

Unit relation:

$$(\text{kg/s})(\text{kJ/kg}) = \text{kW}$$

KINETIC ENERGY CHANGE

$$\Delta ke = \frac{V_2^2 - V_1^2}{2}$$

$$\text{m}^2/\text{s}^2 = \text{J/kg}$$

If $V_1 \approx V_2$:

$$\Delta ke \approx 0$$

POTENTIAL ENERGY CHANGE

$$\Delta pe = g(z_2 - z_1)$$

STEADY-FLOW ENERGY EQUATION: SUMMARY

$$\dot{Q} - \dot{W} = \sum_{\text{out}} \dot{m} \left(h + \frac{V^2}{2} + gz \right) - \sum_{\text{in}} \dot{m} \left(h + \frac{V^2}{2} + gz \right)$$

Single stream:

$$\dot{Q} - \dot{W} = \dot{m} \left[h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1) \right]$$

Per unit mass:

$$q - \omega = h_2 - h_1 + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)$$

If $\Delta ke, \Delta pe \approx 0$:

$$q - \omega = h_2 - h_1$$

5.4 SOME STEADY-FLOW ENGINEERING DEVICES

Device Name	Function	Governing Equation	Assumption
Nozzle	Converts enthalpy into kinetic energy; velocity increases.	$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$	$\dot{Q} \approx 0, \dot{W} = 0, \Delta pe \approx 0$
Diffuser	Converts kinetic energy into enthalpy; velocity decreases.	$h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$	$\dot{Q} \approx 0, \dot{W} = 0, \Delta pe \approx 0$
Turbine	Produces shaft work from the flowing fluid.	$w_{out} = h_1 - h_2$	$\dot{Q} \approx 0, \Delta ke \approx 0, \Delta pe \approx 0$
Compressor	Raises the pressure of a gas by consuming shaft work.	$w_{in} = h_2 - h_1 - q$ Adiabatic: $w_{in} = h_2 - h_1$	$\dot{Q} \approx 0, \Delta ke \approx 0, \Delta pe \approx 0$
Pump	Raises the pressure of a liquid by consuming shaft work.	$w_{in} = h_2 - h_1 - q$ Adiabatic: $w_{in} = h_2 - h_1$	$\dot{Q} \approx 0, \Delta ke \approx 0, \Delta pe \approx 0$
Fan	Produces a relatively small pressure increase in a gas.	$w_{in} = h_2 - h_1 - q$	$\dot{Q} \approx 0, \Delta ke \approx 0, \Delta pe \approx 0$

Device Name	Function	Governing Equation	Assumption
Throttling Valve	Produces a large pressure drop with no useful work output; isenthalpic device.	$h_2 \approx h_1$ $u_1 + P_1 v_1 = u_2 + P_2 v_2$	$q \approx 0, \quad \dot{w} = 0,$ $\Delta ke \approx 0,$ $\Delta pe \approx 0$
Mixing Chamber	Mixes two or more streams; mass and energy are conserved.	$\sum_{in} \dot{m} = \sum_{out} \dot{m}$ $\sum_{in} \dot{m} h_{in} = \sum_{out} \dot{m} h_{out}$	$\dot{Q} \approx 0, \quad \dot{W} = 0,$ $\Delta ke \approx 0,$ $\Delta pe \approx 0$
Heat Exchanger	Transfers heat between streams without mixing them.	$\sum_{in} \dot{m} h = \sum_{out} \dot{m} h$ $\dot{Q}_{gained} = \dot{Q}_{lost}$	$\dot{W} = 0,$ $\Delta ke \approx 0,$ $\Delta pe \approx 0,$ $\dot{Q}_{CV} \approx 0$
Pipe / Duct	Transports fluid; heat transfer and work interactions may occur.	$\dot{Q} - \dot{W} = \dot{m}(h_2 - h_1)$ $\dot{Q} - \dot{W} = \dot{m} c_p (T_2 - T_1)$	Steady flow, $\Delta ke \approx 0,$ $\Delta pe \approx 0$

5.5 ENERGY ANALYSIS OF UNSTEADY-FLOW PROCESSES

CHARACTERISTICS OF UNSTEADY-FLOW SYSTEMS

Steady flow:

$$\frac{dm_{CV}}{dt} = 0, \quad \frac{dE_{CV}}{dt} = 0$$

Unsteady flow:

Mass and/or energy within the control volume may change with time.

The CV is usually stationary, but its boundary may move; therefore boundary work can occur.

$$\text{Unsteady flow} \neq \text{steady flow}$$

MASS BALANCE FOR AN UNSTEADY-FLOW PROCESS

$$m_{in} - m_{out} = \Delta m_{system}$$

$$\Delta m_{system} = m_{final} - m_{initial}$$

Thus,

$$m_{in} - m_{out} = m_{final} - m_{initial}$$

For a control volume:

$$m_i - m_e = (m_2 - m_1)_{CV}$$

Useful special cases:

$$m_i = 0 \quad \text{no inlet}$$

$$m_e = 0 \quad \text{no exit}$$

$$m_1 = 0 \quad \text{initially evacuated CV}$$

ENERGY CHANGES IN AN UNSTEADY-FLOW PROCESS

$$E_{in} - E_{out} = \Delta E_{system}$$

$$\text{Net energy entering} = \text{Change in energy stored}$$

Energy may cross the boundary through:

$$\text{heat} + \text{work} + \text{mass flow}$$

UNIFORM-FLOW PROCESS

A uniform-flow process is an idealization used for practical unsteady-flow processes. The inlet/exit properties are treated as uniform and steady during the process, although the control volume itself may change state with time.

$$\text{Uniform flow} \neq \text{steady flow}$$

The state of the exiting mass at an instant is taken to correspond to the state of fluid within the CV at that instant.

ENERGY OF FLOWING AND NONFLOWING FLUID

For flowing fluid:

$$\theta = h + ke + pe$$

$$ke = \frac{V^2}{2} \quad pe = gz$$

Therefore,

$$\theta = h + \frac{V^2}{2} + gz$$

For nonflowing fluid within the CV:

$$e = u + ke + pe$$

$$e = u + \frac{V^2}{2} + gz$$

Key distinction:

$$\text{Flowing fluid} \rightarrow h \quad \text{Nonflowing fluid} \rightarrow u$$

GENERAL ENERGY EQUATION FOR A UNIFORM-FLOW PROCESS

$$\left(Q_{\text{in}} + W_{\text{in}} + \sum_{\text{in}} m\theta \right) - \left(Q_{\text{out}} + W_{\text{out}} + \sum_{\text{out}} m\theta \right) = (m_2 e_2 - m_1 e_1)_{\text{system}}$$

with

$$\theta = h + \frac{V^2}{2} + gz$$

$$e = u + \frac{V^2}{2} + gz$$

NEGLIGIBLE KINETIC AND POTENTIAL ENERGY CHANGES

If

$$\Delta ke \approx 0, \quad \Delta pe \approx 0$$

then

$$Q - W = \sum_{\text{out}} mh - \sum_{\text{in}} mh + (m_2 u_2 - m_1 u_1)_{\text{system}}$$

where

$$Q = Q_{\text{net,in}} = Q_{\text{in}} - Q_{\text{out}}$$

$$W = W_{\text{net,out}} = W_{\text{out}} - W_{\text{in}}$$

PHYSICAL MEANING OF THE SIMPLIFIED ENERGY EQUATION

$$Q - W = \sum_{\text{out}} mh - \sum_{\text{in}} mh + (m_2 u_2 - m_1 u_1)_{\text{system}}$$

Unlike a steady-flow device, an unsteady-flow CV can store or release energy as its internal state changes.

$$\text{Energy in} = \text{energy out} + \text{energy accumulation}$$

CHARGING OF A RIGID TANK

For one inlet, no outlet:

$$m_i = m_2 - m_1$$

Initially evacuated:

$$m_1 = 0 \quad \Rightarrow \quad m_i = m_2$$

For a rigid, insulated tank with no work and negligible kinetic/potential-energy changes:

$$Q = 0, \quad W = 0$$

Energy balance:

$$m_i h_i = m_2 u_2 - m_1 u_1$$

For an initially evacuated tank:

$$m_i h_i = m_2 u_2$$

Using

$$m_i = m_2$$

gives

$$u_2 = h_i$$

Since

$$h = u + Pv$$

the flow energy Pv entering with the fluid can become internal energy after the fluid comes to rest inside the tank.

ALTERNATIVE CLOSED-SYSTEM INTERPRETATION OF TANK CHARGING

Boundary work for the imaginary closed-system representation:

$$W_{b,in} = - \int_1^2 P_i dV$$

At constant pressure:

$$W_{b,in} = -P_i(V_2 - V_1)$$

The resulting energy relation is

$$u_2 = u_i + P_i v_i$$

Since

$$h_i = u_i + P_i v_i$$

$$u_2 = h_i$$

Thus the closed-system and uniform-flow interpretations give the same result.

DISCHARGING OF A TANK

For one exit, no inlet:

$$m_e = m_1 - m_2$$

For a rigid tank:

$$\dot{W}_b = 0$$

For an insulated tank:

$$\dot{Q} \approx 0$$

With negligible kinetic and potential-energy changes:

$$\dot{W}_{in} - \dot{m}_e h_e = \dot{m}_2 u_2 - \dot{m}_1 u_1$$

Electrical work supplied to the tank can compensate for the energy carried away by the discharged fluid.

CONSTANT-TEMPERATURE DISCHARGE

For constant temperature:

$$T_2 = T_1$$

For a fluid for which $u = u(T)$:

$$u_2 = u_1$$

The process may have no accumulation at a particular state while mass and energy inside the CV still vary with time:

$$\text{No accumulation within CV}$$

$$\text{Mass and energy within CV may change with time}$$

The essential unsteady-flow interactions are:

$$\text{Mass transfer} + \text{Energy transfer} + \text{Energy storage}$$